MD SAIF ALVI

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Education

Khulna University of Engineering and Technology *Khulna, Bangladesh*
BS, Electronics and Communication Engineering *Jan 2020 – Sep 2025*
GPA: 3.41/4.00 (1 year delay due to Covid)

Research Experience

RISE Research Group 2026
Advisor: Md Ahsan Kabir

- Contributed to annotating and curating a 930-image industrial fastener dataset, converting polygon annotations into YOLO bounding boxes, and auditing data quality.
- Implemented reproducible comparisons of YOLOv8m, YOLO11m, YOLO12m, and YOLO26m locally, evaluating Clip, Rivet, and Screw detection with validation metrics.

Undergraduate Thesis 2025
Dept. of Electronics and Communication Engineering, Khulna University of Engineering and Technology
Advisor: Md. Khorshed Alom
External advisor: Dr. Monir Hossen

- Demonstrated that integrating DBSCAN clustering, synchronized dual-side GRU prediction, and sleep-awake scheduling can extend the network lifetime of Wireless Sensor Networks (WSNs) to over 20,000 rounds while achieving an 82.24% transmission avoidance ratio and a 100% Packet Delivery Ratio (PDR).
- Examined the framework using network stability metrics (First Node Death, Half Node Death, Last Node Death), total energy consumption, average latency, and Transmission Avoidance Ratio (TAR) on a simulated 100×100 m homogeneous sensor field against LSTM, LMS-LSTM, and un-scheduled GRU benchmarks.
- Prepared materials for a peer-reviewed conference manuscript published in the 2026 International Conference on Computing, Communication, Security and Intelligent Systems (IC3SIS).

Publications

- **MS Alvi,** RT Joita, MK Alom, "DGS-WSN: A DBSCAN Enabled GRU Based Sleep Scheduling Framework for Energy Efficient Data Transmission in WSNs," IEEE International Conference on Computing, Communication, Security and Intelligent Systems (IC3SIS), Kerala, India, 2026, doi: 10.1109/IC3SIS69949.2026.11608327.
- MAK Kausik, TA Refat, **MS Alvi,** TB Ovi, MA Kabir, "Toward Zero Missed Detections in Low-Data Industrial Fastener Inspection: A Comparative Study of YOLOv11 and YOLOv12," submitted to International Journal of Engineering Science. (*Under review*)

Professional Experience

Software Engineer Jul 2025 - Present
Apploye Inc., Dhaka, Bangladesh
Linux, Shell, Django, Typescript, REST, GraphQL, React, Javascript, PostgreSQL, Payload CMS, Next.js, Shopify, RabbitMQ, BullMQ

- Led full-stack engineering of a lead-collection platform spanning four codebases, modernizing its payment infrastructure, third-party integrations, and UI while the user base increased by 80%.
- Redesigned dashboard APIs, Payment webhooks and system components to improve architecture and efficiency, achieving at least 30% lower API response times.
- Developed a Shopify application, from requirements analysis and system architecture to database design and full-stack implementation, including a fully customizable canvas and 10+ third-party integrations.

- Architected asynchronous worker infrastructure using BullMQ for integrations, scheduling, and background processing, establishing reliable execution of long-running and event-driven tasks.
- Built two production websites from the ground up using Payload CMS, establishing reusable content-management workflows that enabled non-technical teams to manage content independently.

Apprenticeship Dec 2024 - Apr 2025
Geeky Solutions, Remote
Django REST Framework, React, PostgreSQL, GitHub Actions, Pytest

- Under the supervision of a senior engineer, collaborated in a 3-member team to design and implement a full-stack learning management system.
- Led the development of a timed assessment system with dynamic question generation, result analytics, and grading logic.
- Achieved a SonarQube code quality score of 96/100 (3rd highest among 292 teams) and configured the CI/CD pipeline successfully.
- Implemented unit tests, validated APIs and documented the API endpoints to improve reliability.

Selected Projects

- KUETx: Learning Management System. 2025
  Hackathon, Collaborated in a team of 3 members. GitHub
- Wireless Digital Communication System Design under Noise. (MATLAB Simulink) 2024
- Campus-Scale Multi-VLAN Network. (CISCO Packet Tracer) 2024
- Arduino–nRF24L01+PA+LNA based Walkie-Talkie Prototype.
  Advisor: Dr. Monir Hossen 2023
- Inset-Fed Triangular Patch Antenna. (CST Suite) 2023
- FIR-Based Audio Noise Filter. (MATLAB) 2023
- Full-Stack E-commerce Platform Development. (PHP) 2023
- Employee Management System. (Oracle DBMS) 2022
- Regulated DC Power Supply Prototype. 2022

Achievements

- Runner-up, Learnathon 3.0 by Geeky Solutions. 2025
  2nd position among 292 registered teams, 3rd highest code quality score.
- Solved 1000+ algorithmic problems in C++. 2020-2025
  Platforms: Codeforces, AtCoder, and LeetCode.
- Technical scholarship: Merit based award. 2021-2025
  For 4 consecutive academic years.
- 2nd runner-up (Tech), AI hackathon, Apploye Inc. 2025

References

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Dr. Monir Hossen, SMIEEE, FIEB
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